Revision 2021 • Program Core • Semester 4

Fluid Mechanics & Hydraulic Machinery

A complete diploma engineering handbook for course code 4022. It covers fluid properties, pressure measurement, continuity, Bernoulli theorem, flow meters, notches, orifices, pipe losses, impact of jets, hydraulic turbines and hydraulic pumps.

> □□□ short note □□□□. Pressure head, manometer, Bernoulli, pipe losses, turbine power, pump efficiency □□□□□□ diagram + formula + numerical method □□□□□ □□□□□□□□ □□□□□□□□□□□□□ handbook □□□.

Bernoulli Head Balance

Pressure head + Velocity head + Datum head

Energy view used in pipe, meter, turbine and pump problems.

Course Overview, Source Information and Roadmap

What this subject is

Fluid Mechanics & Hydraulic Machinery studies behaviour of fluids at rest and in motion, pressure measurement, energy conservation, pipe losses, turbines and pumps. It is directly useful in water supply, irrigation, hydroelectric plants, workshop test rigs, pump selection, maintenance and fault diagnosis.

> Pipe diameter change □□□□□□□□ velocity □□□□□□. Pump head □□□□□□□□ motor power □□□□□□□ □□□□□□. Cavitation □□□□□□□ pump/turbine damage □□□□□□□□. □ practical understanding □□□ subject-□□□□ value.

Official information

Program: Diploma in Mechanical Engineering / Manufacturing Technology
Course code: 4022
Semester: 4
Credits: 4
Category: Program Core
Periods: 4 per week, 60 per semester

Suggested practicals: Not specified in supplied syllabus/source. Model paper pattern: Not specified in supplied syllabus/source.

Course outcomes and CO-PO mapping

CO	Outcome	Hours	Level	Mapping
CO1	Explain fluid properties and pressure measurement techniques.	12	Understanding	PO1=3, PO2=3, PO7=1
CO2	Apply conservation laws to fluid flow over notches and through pipes and orifices.	17	Applying	PO1=3, PO2=3, PO7=1
CO3	Describe construction, working and performance testing of hydraulic turbines.	17	Understanding	PO1=3, PO2=3, PO7=1
CO4	Describe construction, working and performance testing of hydraulic pumps.	12	Understanding	PO1=3, PO2=3, PO7=1

Redesigned syllabus roadmap

Module 1

Core: fluid properties, pressure, Pascal law, piezometer, manometers, Bourdon gauge.

Draw: manometer, pressure scale, Bourdon gauge. **Calculate:** density, pressure head, manometer readings.

Module 2

Core: flow types, continuity, Bernoulli, Venturi, orifice meter, Pitot tube, notches, pipe losses, HGL/TEL.

Calculate: discharge, velocity, head loss, notch flow.

Module 3

Core: impact of jets, hydroelectric plant, turbine classification, Pelton, Francis, Kaplan, draft tube, cavitation.

Calculate: work done, power, efficiency, unit quantities.

Module 4

Core: centrifugal pumps, priming, cavitation, manometric head, reciprocating pump, slip, hydraulic ram and special pumps.

Calculate: pump power, efficiencies, discharge and slip.

Module 1: Fluid Properties and Pressure Measurement

Outcome: Explain fluid properties and pressure measurement techniques. Duration: 12 hours.

Textbook chapter

Properties of fluids

Density is mass per unit volume. Specific weight is weight per unit volume. Specific volume is volume per unit mass. Specific gravity is density ratio with water and has no unit. Viscosity is resistance to relative motion between layers. Surface tension and capillarity are important in small tubes. Vapour pressure is linked with cavitation.

Pressure and pressure head

Pressure is force per unit area. Pressure head expresses pressure as height of fluid column: h = p/(rho g). Absolute pressure is measured from perfect vacuum. Gauge pressure is measured from atmospheric pressure. Vacuum pressure means pressure lower than atmosphere.

Pascal law

Pressure applied to a confined fluid is transmitted equally in all directions. Hydraulic press, hydraulic lift and hydraulic brake are based on this principle.

Pressure measuring instruments

Piezometer measures low positive pressure using liquid column. U-tube and differential manometers measure pressure difference using liquid columns. Bourdon gauge converts pressure into elastic tube movement.

Important formulas

```
rho = m/V

gamma = rho g

SG = rho fluid / rho water

p = F/A
```

```
h = p/(rho g)
```

Diagram checklist

- Fluid property map
- Pressure scale
- Hydraulic press for Pascal law
- Piezometer
- U-tube and differential manometer
- Bourdon gauge

U-tube manometer balances pressure by liquid column height

Manometer principle

Expected questions

1. Define density, specific weight, specific volume and specific gravity.
2. Explain viscosity, surface tension, capillarity, vapour pressure and compressibility.
3. State and explain Pascal law.
4. Explain absolute, gauge, atmospheric and vacuum pressure.
5. Explain piezometer, manometer and Bourdon gauge.
6. Solve problems on pressure head and manometer.

Module 2: Fluid Flow, Bernoulli, Notches, Orifices and Pipe Losses

Outcome: Apply conservation laws to fluid flow over notches and through pipes and orifices. Duration: 17 hours.

Textbook chapter

Types of flow

Flow may be steady/unsteady, uniform/non-uniform, compressible/incompressible, rotational/irrotational and one-dimensional/multidimensional. Discharge is volume flow rate, Q = A V. Streamline is tangent to velocity direction; path line is actual particle path.

Continuity and Bernoulli

Continuity equation A1V1 = A2V2 expresses conservation of mass for steady incompressible flow. Bernoulli equation p/gamma + V2/2g + z = constant represents conservation of mechanical energy along a streamline.

Flow meters and notches

Venturimeter, orifice meter and Pitot tube use pressure changes to measure flow. Rectangular and triangular notches measure open channel discharge. Orifices are openings in tank walls/plates and use hydraulic coefficients.

Pipe losses

Major loss is friction over pipe length. Minor losses occur at entry, exit, bends, valves, sudden enlargement and contraction. HGL shows pressure plus datum head; TEL also includes velocity head.

Key formulas

```
Q = A V

p/gamma + V²/2g + z = constant

Q rectangular notch = (2/3) Cd b sqrt(2g) H^(3/2)

hf = 4 f L V²/(2 g d)
```

Common mistakes

- Using diameter instead of area in continuity.
- Forgetting coefficient of discharge.
- Using Bernoulli without stating limitations.
- Mixing major and minor losses.

HGL and TEL fall in direction of flow due to head loss

Hydraulic and total energy lines

Expected questions

1. Classify fluid flow with examples.
2. State continuity equation and solve problems.
3. Derive/state Bernoulli equation and limitations.
4. Explain Venturimeter, orifice meter and Pitot tube.
5. Explain notches and orifices with discharge equations.
6. Explain Darcy and Chezy equations, HGL and TEL.
7. Solve major and minor loss numericals.

Module 3: Impact of Jets and Hydraulic Turbines

Outcome: Describe construction, working and performance testing of hydraulic turbines. Duration: 17 hours.

Textbook chapter

Impact of jet

A water jet has momentum. When it strikes a plate or curved vane, change in momentum creates force. Turbine buckets/blades are shaped to change water momentum and receive work.

Hydroelectric plant

Main parts are reservoir, dam, intake, trash rack, penstock, surge tank, turbine, draft tube, generator, tail race and switchyard. Water potential energy becomes mechanical and then electrical energy.

Turbine classification

Turbines are classified as impulse/reaction, tangential/radial/axial/mixed flow and by head. Pelton suits high head low discharge, Francis suits medium head, Kaplan suits low head high discharge.

Draft tube and cavitation

Draft tube recovers kinetic energy at reaction turbine exit. Cavitation occurs when pressure falls below vapour pressure and vapour bubbles collapse, causing noise, vibration and pitting.

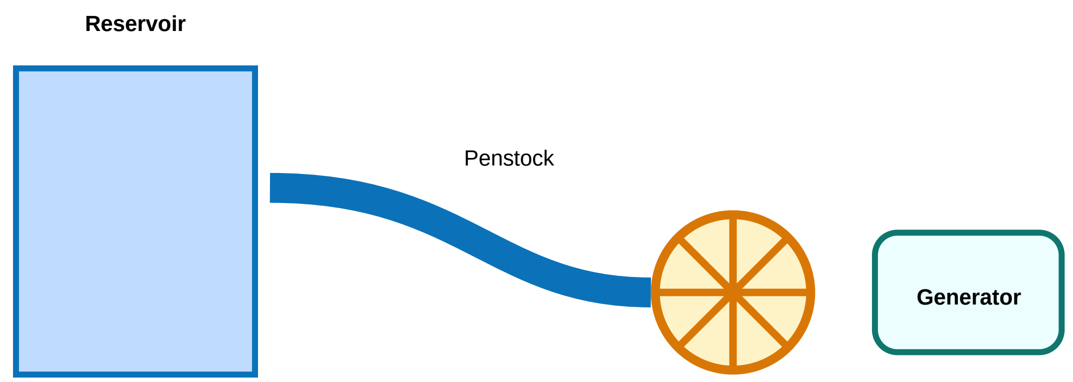
Formulas

```
Hydraulic power = rho g Q H

Efficiency = output power / input water power

Unit quantities are used for comparing turbines under unit head.
```

What to draw

- Impact of jet on plate/vane
- Hydroelectric plant layout

- Pelton wheel with nozzle
- Francis and Kaplan turbine
- Draft tube



Hydroelectric power plant layout

Expected questions

1. Explain impact of jet on fixed and moving plates.
2. Explain impact of jet on curved vane.
3. Draw hydroelectric power plant layout.
4. Classify turbines and select turbine based on head/discharge.
5. Explain Pelton, Francis and Kaplan turbines.
6. Explain draft tube and cavitation.
7. Solve turbine power and efficiency problems.

Module 4: Hydraulic Pumps

Outcome: Describe construction, working and performance testing of hydraulic pumps. Duration: 12 hours.

Textbook chapter

Centrifugal pump

A rotating impeller gives kinetic energy to liquid. The casing converts part of kinetic energy into pressure energy. Main parts are casing, impeller, suction pipe, foot valve, strainer, delivery pipe and shaft.

Priming and cavitation

Priming is filling casing and suction pipe with liquid before starting. Cavitation happens when local pressure drops below vapour pressure. It causes noise, vibration, pitting and loss of efficiency.

Reciprocating pump

A piston/plunger moves to create suction and delivery. Single acting pump delivers during one stroke; double acting pump delivers during both strokes. It is positive displacement and used for high pressure low discharge.

Special pumps

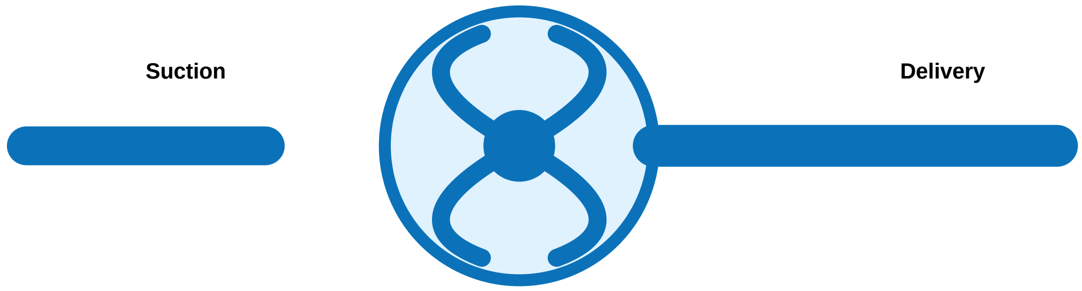
Hydraulic ram uses water hammer to lift water. Air lift pump uses compressed air. Jet pump uses high velocity jet. Submersible, propeller and turbine pumps are used for specific discharge/head requirements.

Formulas

```
Water power = rho g Q H

Overall efficiency = water power / shaft power

Q single acting = A L N / 60

Slip = Qth - Qact
```

Common mistakes

- Starting centrifugal pump without priming.
- Confusing cavitation and separation.
- Using wrong discharge formula for double acting pump.
- Forgetting to convert L/s into m^3/s.



Centrifugal pump: impeller gives velocity, casing converts to pressure

Centrifugal pump principle

Expected questions

1. Explain centrifugal pump construction and working.
2. Explain priming and methods.
3. Explain cavitation in pumps and prevention.
4. Define manometric head and efficiencies.
5. Explain single/double acting reciprocating pump.
6. Define slip and negative slip.
7. Explain hydraulic ram and special pumps.
8. Solve pump power/discharge numericals.

Formula / Rule Bank

Density

```
rho = m / V
```

Unit kg/m^3. Used to find mass or property of fluids.

Specific weight

```
gamma = rho g
```

Unit N/m³. Used in pressure head relation.

Specific gravity

```
SG = rho fluid / rho water
```

No unit. Density of fluid = SG x 1000 for water reference.

Pressure

```
p = F / A
```

Unit Pa. Used in pressure intensity calculations.

Pressure head

```
h = p / (rho g)
```

Head in metres of fluid.

Continuity

```
A1 V1 = A2 V2 = Q
```

Conservation of mass for steady incompressible flow.

Bernoulli

```
p/gamma + V²/2g + z = constant
```

Energy equation along streamline.

Darcy loss

```
hf = 4 f L V² / (2 g d)
```

Major friction loss in pipe.

Minor loss

```
hm = K V² / 2g
```

Loss due to fitting, valve, bend, entry or exit.

Hydraulic power

```
P = rho g Q H
```

Water power for pump/turbine.

Pump efficiency

```
eta = water power / shaft power
```

Use same units before ratio.

Reciprocating pump

```
Q = A L N / 60
```

Single acting theoretical discharge.

Solved Examples / Problem Lab

1. Density

A liquid of mass 600 kg occupies 0.75 m^3. Density = 600/0.75 = 800 kg/m^3. Specific weight = 800 x 9.81 = 7848 N/m^3.

2. Pressure head

Pressure = 196.2 kPa. h = p/(rho g) = 196200/(1000 x 9.81) = 20 m of water.

3. Continuity

$A1=0.04\ m^2$, $A2=0.02\ m^2$, V1=1.5 m/s. V2=A1V1/A2 = 3 m/s.

4. Bernoulli

Use $p1/gamma+V1^2/2g+z1 = p2/gamma+V2^2/2g+z2$. Substitute every value as metre head, then solve unknown pressure head.

5. Darcy loss

L=80 m, d=0.1 m, V=2 m/s, f=0.005. $hf=4fLV^2/(2gd)=3.26$ m approximately.

6. Pump efficiency

$Q=0.04\ m^3/s$, H=25 m, shaft power=15 kW. Water power=1000x9.81x0.04x25=9.81 kW. Efficiency=9.81/15=65.4%.

Practice and Quick Revision

Last-day revision

- CO1: properties, pressures and manometers.
- CO2: continuity, Bernoulli, meters, notches, losses.
- CO3: jets, hydro plant, turbines, draft tube.
- CO4: pumps, priming, cavitation, slip.

Exam traps

- Specific gravity has no unit.
- Pressure head is in metres.
- Bernoulli needs assumptions.
- Diagrams must be labelled.

- Always convert litres to m^3.

Objective questions

1. Density is

A. weight/volume B. mass/volume C. volume/mass D. force/area

Answer: B

2. Venturimeter works on

Bernoulli theorem.

3. Pelton turbine is suited for

High head and low discharge.

4. Priming is required in

Centrifugal pump.

5. Cavitation occurs when

Local pressure falls below vapour pressure.

Fresh Model Question Paper

Time: 3 Hours | Maximum Marks: 100. Official model paper pattern is not specified in supplied syllabus/source, so this is a fresh diploma-style practice paper.

Part A - Short Answer (10 x 2 = 20)

1. Define density and specific gravity.
2. What is pressure head?
3. State Pascal law.
4. List two pressure measuring instruments.
5. State continuity equation.
6. Mention two applications of Bernoulli theorem.
7. What is cavitation?
8. What is priming?
9. Define slip.
10. Name turbines suitable for high, medium and low head.

Part B - Analytical Answers (8 x 5 = 40)

1. Explain absolute, gauge and vacuum pressures.
2. Explain U-tube manometer with sketch.
3. Classify fluid flow.
4. Derive/state Bernoulli equation and limitations.
5. Explain Venturimeter.
6. Explain pipe losses and HGL/TEL.
7. Explain impact of jet on plates and vanes.
8. Explain centrifugal pump construction and working.

Part C - Descriptive / Numerical (4 x 10 = 40)

1. A liquid has mass 720 kg and volume 0.9 m^3. Find density, specific weight and SG.
2. A pipe reduces from 200 mm to 100 mm diameter. Velocity in larger pipe is 1.2 m/s. Find smaller pipe velocity and discharge.
3. Draw hydroelectric power plant and explain turbine classification/selection.
4. A pump delivers 0.05 m^3/s against 30 m head. Shaft power is 22 kW. Find water power and efficiency.

Complete Model Answers

Numerical C1

rho=m/V=720/0.9=800 kg/m^3. gamma=rho g=800x9.81=7848 N/m^3. SG=800/1000=0.8.

Numerical C2

$A1/A2=(D1/D2)^2=(0.2/0.1)^2=4$. $V2=4x1.2=4.8$ m/s. $Q=A1V1=\pi/4 \times 0.2^2 \times 1.2=0.0377$ m^3/s.

Numerical C4

Water power=rho g Q H=1000x9.81x0.05x30=14.715 kW. Efficiency=14.715/22=0.669=66.9%.

Diagram answers

For manometer, Venturimeter, hydro plant, turbine and pump questions, draw neat labelled diagram first. Then explain signal/energy/flow path step by step and add formula/application.

References

- Fluid Mechanics and Hydraulic Machines - Dr. R. K. Bansal.
- Hydraulic, Fluid Mechanics and Hydraulic Machines - R. S. Khurmi.
- Hydraulics and Fluid Mechanics - Dr. P. N. Modi and Dr. S. M. Seth.
- Hydraulics, Fluid Mechanics and Fluid Machines - S. Ramamurtham.
- Fluid Mechanics - Yunus A. Cengel and John M. Cimbala.